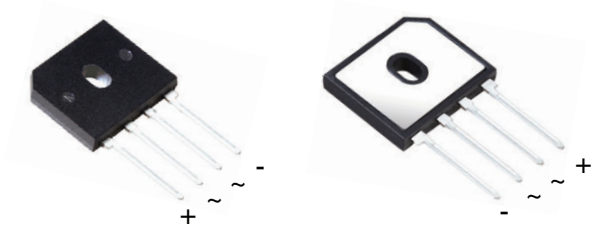
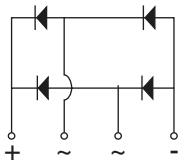
Enhanced isoCink+™ Bridge Rectifiers

isoCink+™



Case Style BU



FEATURES

- UL recognition file number E312394
- Thin single in-line package
- Glass passivated chip junction
- Superior thermal conductivity
- Solder dip 275 °C max. 10 s, per JESD 22-B106
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

RoHS COMPLIANT
HALOGEN FREE Available

TYPICAL APPLICATIONS

General purpose use in AC/DC bridge full wave rectification for switching power supply, home appliances, and white-goods applications.

MECHANICAL DATA

Case: BU

Molding compound meets UL 94 V-0 flammability rating
Base P/N-E3 - RoHS-compliant, commercial grade
Base P/N-M3 - halogen-free, RoHS-compliant, and commercial grade

Terminals: matte tin plated leads, solderable per J-STD-002 and JESD 22-B102
E3 and M3 suffix meet JESD 201 class 1A whisker test

Polarity: as marked on body

Mounting Torque: 10 cm-kg (8.8 inches-lbs) max.

Recommended Torque: 5.7 cm-kg (5 inches-lbs)

LINKS TO ADDITIONAL RESOURCES

3D Models

PRIMARY CHARACTERISTICS	
$I_{F(AV)}$	10 A
V_{RRM}	600 V, 800 V, 1000 V
I_{FSM}	90 A
I_R	5 µA
V_F at I_F = 5.0 A	0.94 V
T_J max.	150 °C
Package	BU
Circuit configurations	In-line

MAXIMUM RATINGS (T_A = 25 °C unless otherwise noted)

PARAMETER		SYMBOL	BU1006A	BU1008A	BU1010A	UNIT
Maximum repetitive peak reverse voltage		V_{RRM}	600	800	1000	V
Average rectified forward current (Fig. 1, 2)	T_C = 90 °C (1)	I_O	10			A
	T_A = 25 °C (2)		3.0			
Non-repetitive peak forward surge current 8.3 ms single sine-wave, T_J = 25 °C		I_{FSM}	90			A
Rating for fusing (t < 8.3 ms) T_J = 25 °C		I^2t	33			A^2s
Operating junction and storage temperature range		T_J, T_{STG}	-55 to +150			°C

Notes

(1) With 60 W air cooled heatsink
(2) Without heatsink, free air

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ELECTRICAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)						
PARAMETER	**TEST CONDITIONS**		**SYMBOL**	**TYP.**	**MAX.**	**UNIT**
Maximum instantaneous forward voltage per diode (1)	I_F = 5.0 A	T_A = 25 °C	V_F	1.02	1.10	V
		T_A = 125 °C		0.94	1.00	
Maximum reverse current per diode	rated V_R	T_A = 25 °C	I_R	-	5.0	µA
		T_A = 125 °C		45	250	
Typical junction capacitance per diode	4.0 V, 1 MHz		C_J	30	-	pF

Note
(1) Pulse test: 300 µs pulse width, 1 % duty cycle

THERMAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)					
PARAMETER	**SYMBOL**	**BU1006A**	**BU1008A**	**BU1010A**	**UNIT**
Typical thermal resistance	$R_{\theta JC}$ (1)	3.0			°C/W
	$R_{\theta JA}$ (2)	20			

Notes
(1) With 60 W air cooled heatsink
(2) Without heatsink, free air

ORDERING INFORMATION (Example)				
PREFERRED P/N	**UNIT WEIGHT (g)**	**PREFERRED PACKAGE CODE**	**BASE QUANTITY**	**DELIVERY MODE**
BU1006A-E3/45	4.48	45	20	Tube
BU1006A-E3/51	4.48	51	250	Paper tray
BU1006A-M3/45	4.48	45	20	Tube

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RATINGS AND CHARACTERISTICS CURVES (T_A = 25 °C unless otherwise specified)

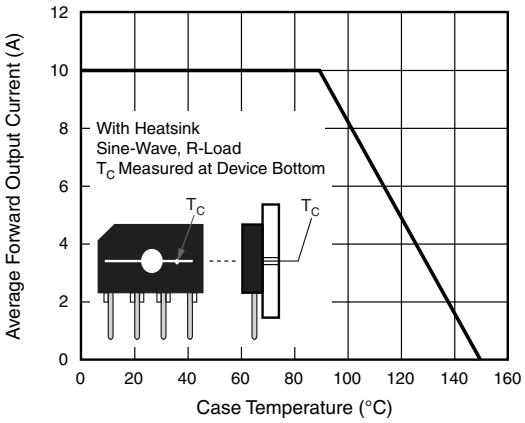


Fig. 1 - Derating Curve Output Rectified Current

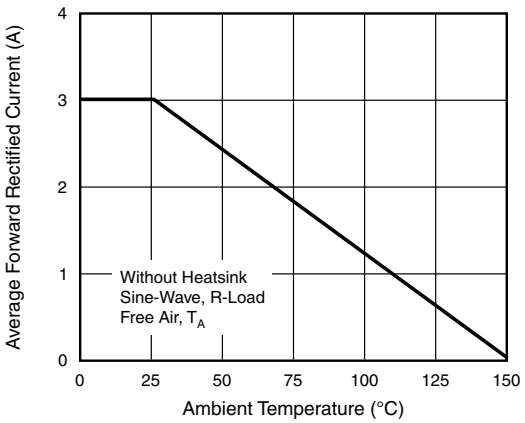


Fig. 2 - Forward Current Derating Curve

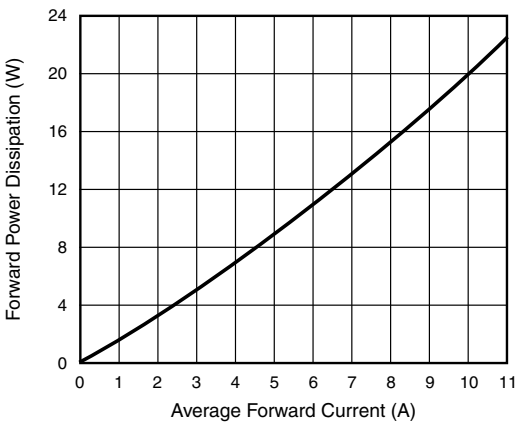


Fig. 3 - Forward Power Dissipation

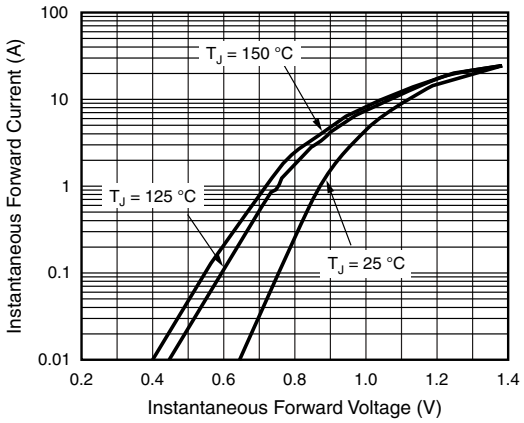


Fig. 4 - Typical Forward Characteristics Per Diode

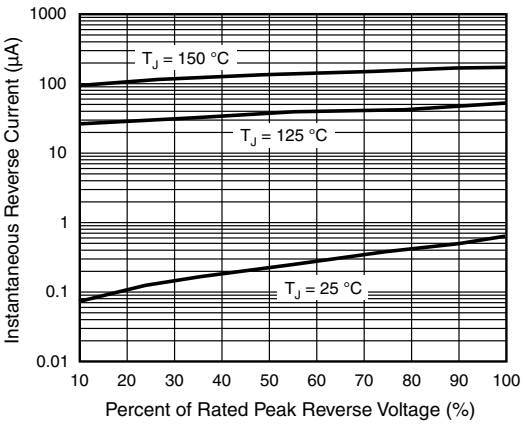


Fig. 5 - Typical Reverse Characteristics Per Diode

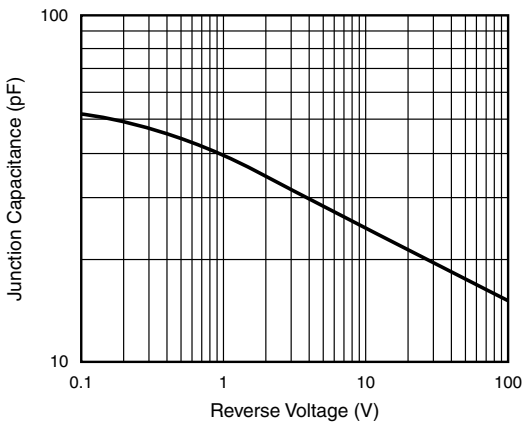
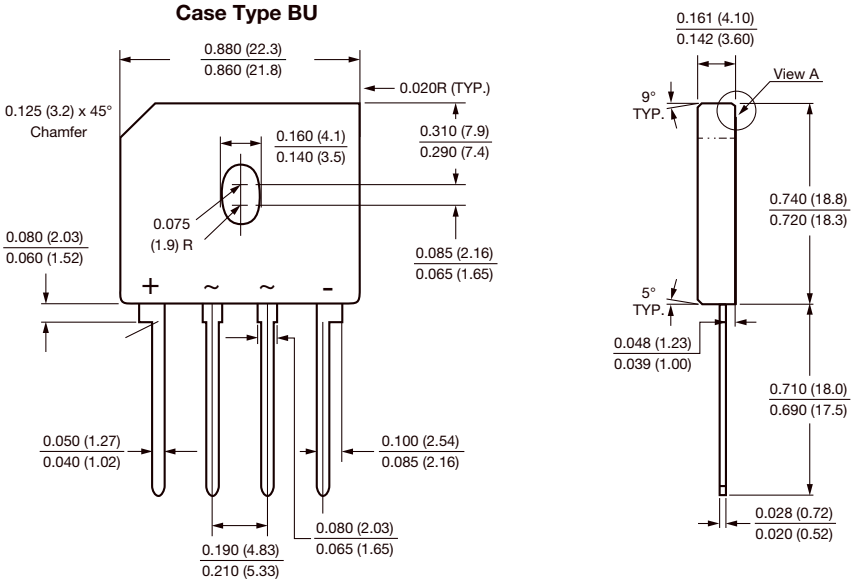


Fig. 6 - Typical Junction Capacitance Per Diode

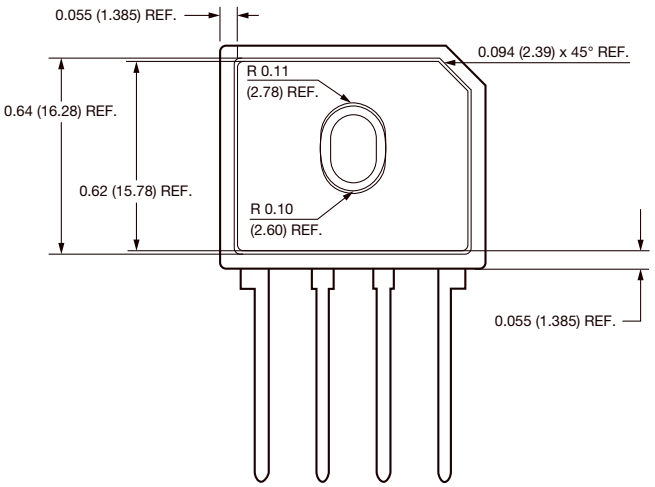
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PACKAGE OUTLINE DIMENSIONS in inches (millimeters)

Case Type BU



Polarity shown on front side of case, positive lead beveled corner

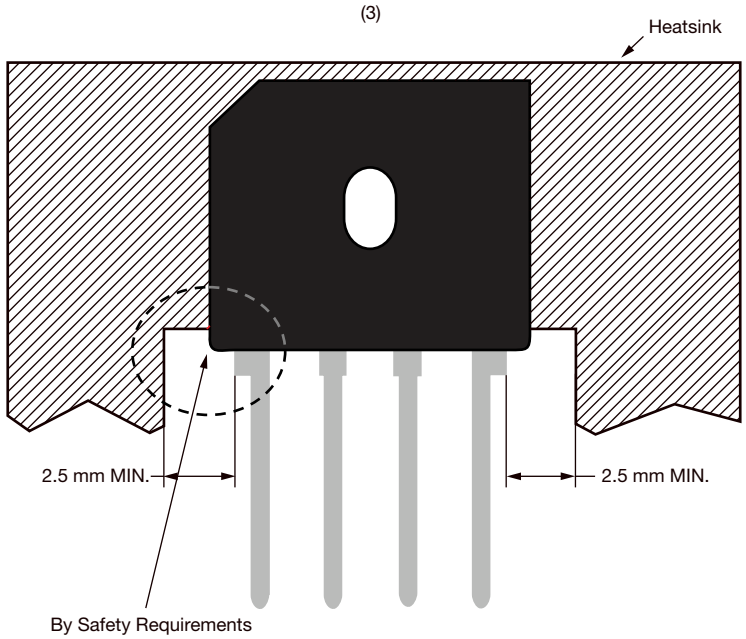


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APPLICATION NOTE

1. Device UL approved for safety use dielectric strength of 1500 V
2. If device is mounted in Floating Ground (F. G.) application, insulator is recommended to use to meet safety requirement.
3. Heat sink shape recommendation:



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